

SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: 5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 2 - Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities. (BL4)

Assessment Tool Weightage: 35%

Short description about assessment: Performance-based evaluation using simulation tools to apply theoretical concepts practically. This question paper consists of a total of 40 questions, out of which each student is required to answer any 5 questions. Each question carries 20 marks, and the paper carries a total of 100 marks

QUESTIONS:5 Total Marks: 100 Duration: 120 Mins

1. Simulation of LTE throughput performance under varying bandwidth, SNR, modulation order, coding rate, and user load (BL4)
2. Simulation of carrier aggregation performance in LTE-Advanced under single carrier, dual carrier, triple carrier, varying bandwidth, and SNR (BL4)
3. Simulation of MIMO channel capacity under antenna number, SNR, bandwidth, fading model, and multiplexing order (BL4)
4. Simulation of beamforming gain under steering angle, antenna elements, frequency, SNR, and interference (BL4)
5. Simulation of 5G throughput performance under QPSK, 16-QAM, 64-QAM, 256-QAM, and varying SNR (BL4)

Code of Conduct:

I Yogeswaran K/192412615 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Simulation Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	20% (4 marks)	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts
Model Design	25% (5 marks)	Develops accurate, well structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model
Tool Usage	20% (4 marks)	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools
Analysis of Results	20% (4 marks)	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results
Documentation	15% (3 marks)	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation

Question 1

Simulation of LTE Throughput Performance under varying Bandwidth, SNR, Modulation Order, Coding Rate, and User Load

Aim

To simulate the throughput performance of an LTE system by varying the Signal-to-Noise Ratio (SNR) and observing the achievable throughput.

Code

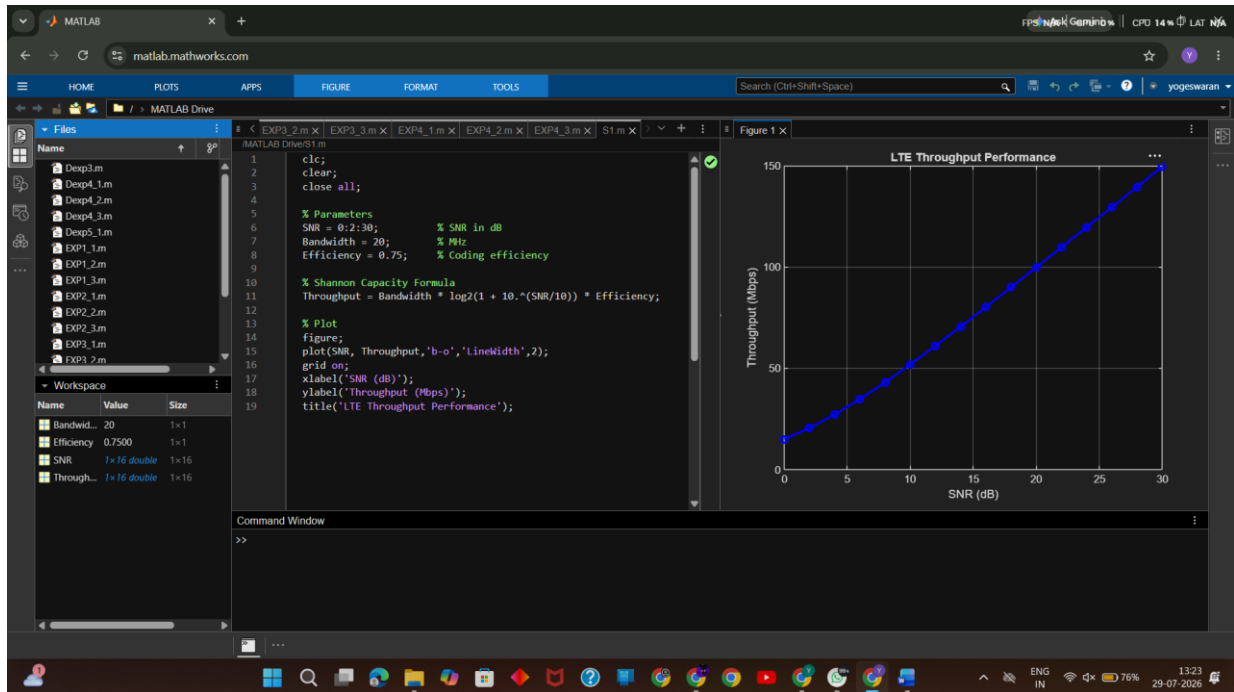
```
clc;
clear;
close all;

% Parameters
SNR = 0:2:30;      % SNR in dB
Bandwidth = 20;    % MHz
Efficiency = 0.75; % Coding efficiency

% Shannon Capacity Formula
Throughput = Bandwidth * log2(1 + 10.^(SNR/10)) * Efficiency;

% Plot
figure;
plot(SNR, Throughput, 'b-o', 'LineWidth', 2);
grid on;
xlabel('SNR (dB)');
ylabel('Throughput (Mbps)');
title('LTE Throughput Performance');
```

Output



Result:

The LTE throughput increases as the SNR increases. At low SNR values, the throughput is relatively low due to poor channel conditions. As SNR improves, the channel capacity increases, resulting in higher data throughput. This demonstrates that better signal quality significantly enhances LTE network performance.

Question 2

Simulation of Carrier Aggregation Performance in LTE-Advanced under Single Carrier, Dual Carrier, Triple Carrier, Varying Bandwidth, and SNR

Aim

To simulate the performance of LTE-Advanced Carrier Aggregation by comparing the throughput of Single Carrier, Dual Carrier, and Triple Carrier configurations under varying SNR conditions.

MATLAB Code

```
clc;
clear;
close all;

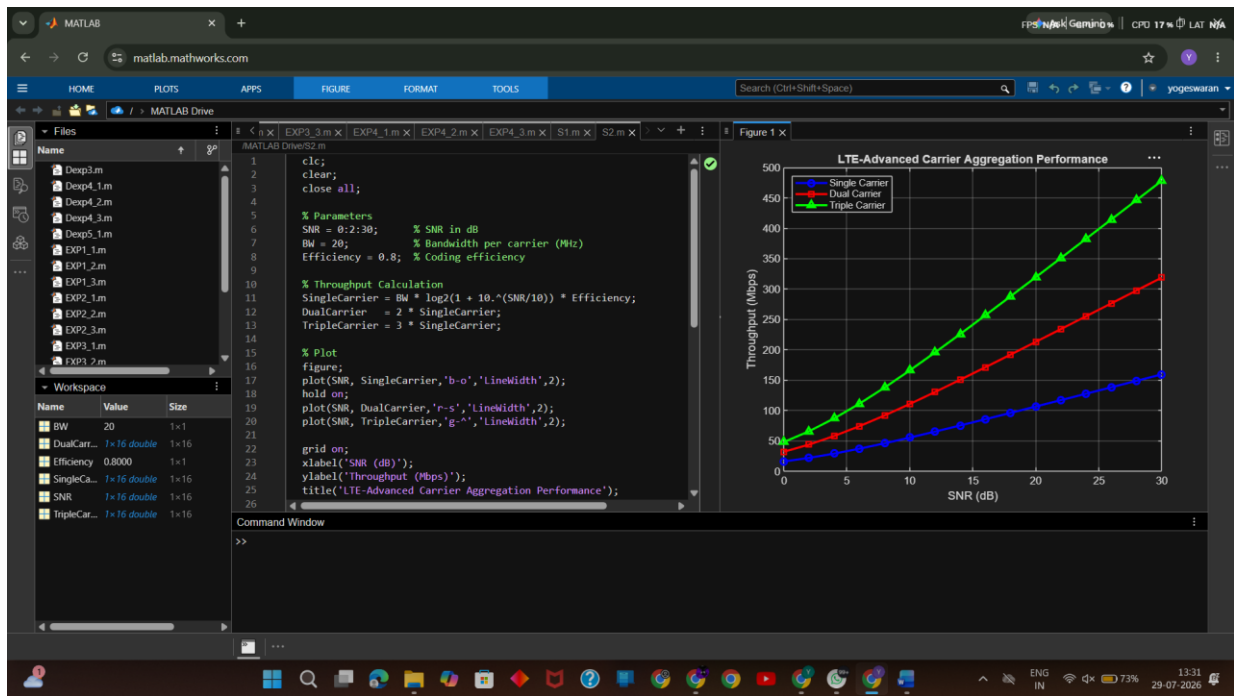
% Parameters
SNR = 0:2:30;    % SNR in dB
BW = 20;         % Bandwidth per carrier (MHz)
Efficiency = 0.8; % Coding efficiency

% Throughput Calculation
SingleCarrier = BW * log2(1 + 10.^(SNR/10)) * Efficiency;
DualCarrier   = 2 * SingleCarrier;
TripleCarrier = 3 * SingleCarrier;

% Plot
figure;
plot(SNR, SingleCarrier, 'b-o', 'LineWidth', 2);
hold on;
plot(SNR, DualCarrier, 'r-s', 'LineWidth', 2);
plot(SNR, TripleCarrier, 'g-^', 'LineWidth', 2);

grid on;
xlabel('SNR (dB)');
ylabel('Throughput (Mbps)');
title('LTE-Advanced Carrier Aggregation Performance');
legend('Single Carrier', 'Dual Carrier', 'Triple Carrier', 'Location', 'northwest');
```

Output



Result:

The simulation shows that throughput increases with increasing SNR for all carrier aggregation configurations. Among the three configurations, **Triple Carrier Aggregation** provides the highest throughput, followed by **Dual Carrier Aggregation** and **Single Carrier**. Carrier aggregation effectively combines multiple frequency bands, resulting in increased bandwidth and improved data transmission performance in LTE-Advanced systems.

Question 3

Simulation of MIMO Channel Capacity under Antenna Number, SNR, Bandwidth, Fading Model, and Multiplexing Order

Aim

To simulate the channel capacity of a MIMO communication system by varying the Signal-to-Noise Ratio (SNR) and comparing the capacity for different antenna configurations.

MATLAB Code

```
clc;
clear;
close all;

% Parameters
SNR = 0:2:30;      % SNR in dB
BW = 20;           % Bandwidth (MHz)

% Antenna Configurations
Nt1 = 2; % 2x2 MIMO
Nt2 = 4; % 4x4 MIMO
Nt3 = 8; % 8x8 MIMO

% Channel Capacity Calculation
Capacity2x2 = BW * Nt1 * log2(1 + 10.^(SNR/10));
Capacity4x4 = BW * Nt2 * log2(1 + 10.^(SNR/10));
Capacity8x8 = BW * Nt3 * log2(1 + 10.^(SNR/10));

% Plot
figure;
plot(SNR, Capacity2x2, 'b-o', 'LineWidth', 2);
hold on;
plot(SNR, Capacity4x4, 'r-s', 'LineWidth', 2);
plot(SNR, Capacity8x8, 'g-^', 'LineWidth', 2);

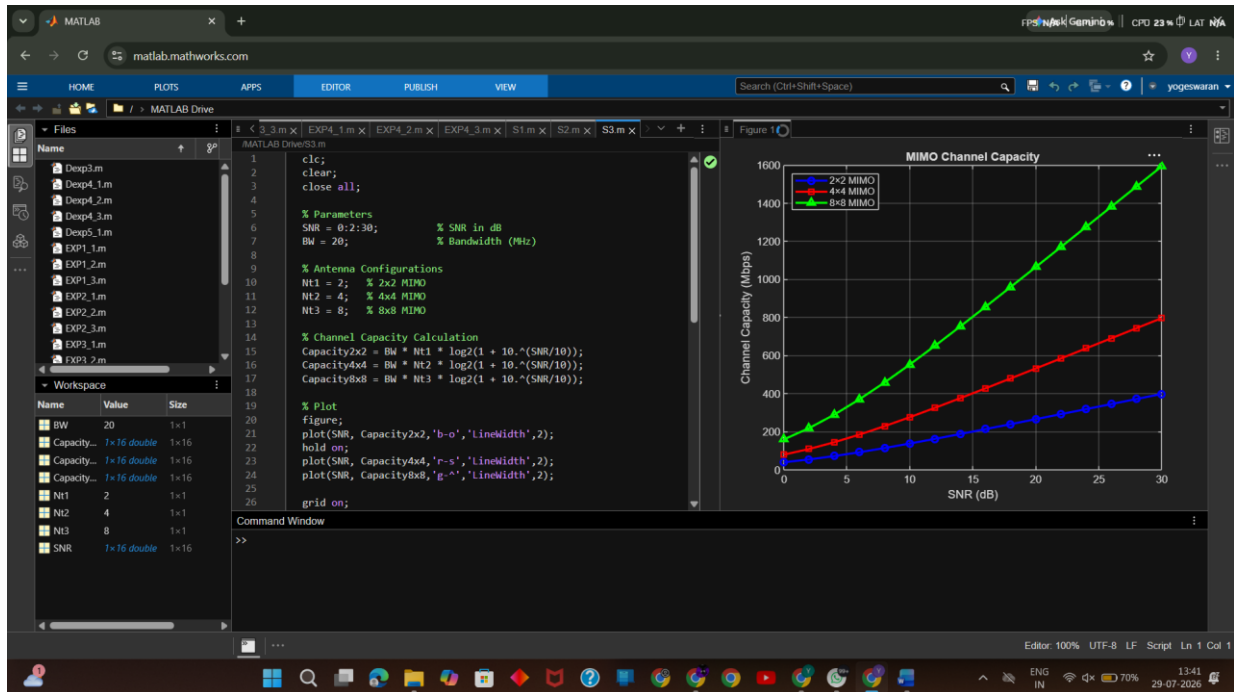
grid on;
```

```

xlabel('SNR (dB)');
ylabel('Channel Capacity (Mbps)');
title('MIMO Channel Capacity');
legend('2×2 MIMO','4×4 MIMO','8×8 MIMO','Location','northwest');

```

Output



Result

The simulation shows that the channel capacity increases with increasing SNR for all MIMO configurations. Among the antenna configurations, 8×8 MIMO achieves the highest channel capacity, followed by 4×4 MIMO and 2×2 MIMO. Increasing the number of antennas improves spatial multiplexing, enabling higher data rates and better spectrum utilization.

Question 4

Simulation of Beamforming Gain under Steering Angle, Antenna Elements, Frequency, SNR, and Interference

Aim

To simulate the beamforming gain of an antenna array by varying the steering angle and observing the corresponding beamforming gain.

MATLAB Code

```
clc;
clear;
close all;

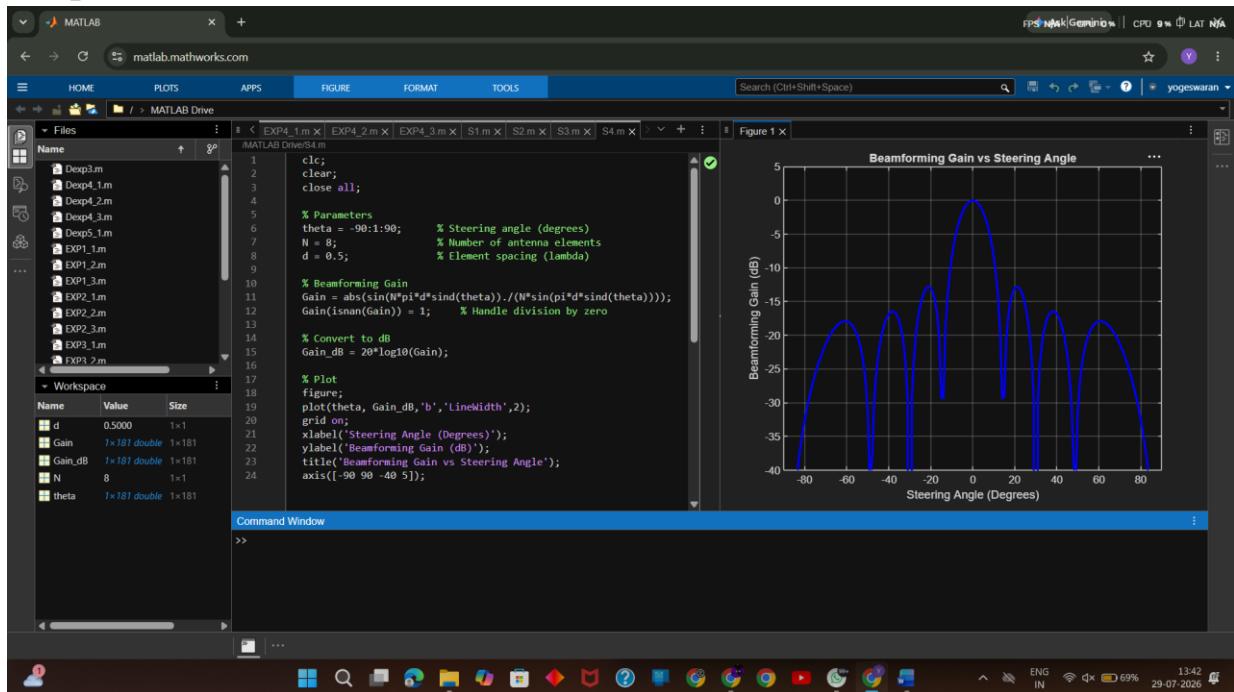
% Parameters
theta = -90:1:90;    % Steering angle (degrees)
N = 8;               % Number of antenna elements
d = 0.5;             % Element spacing (lambda)

% Beamforming Gain
Gain = abs(sin(N*pi*d*sind(theta))./(N*sin(pi*d*sind(theta))));
Gain(isnan(Gain)) = 1; % Handle division by zero

% Convert to dB
Gain_dB = 20*log10(Gain);

% Plot
figure;
plot(theta, Gain_dB, 'b', 'LineWidth', 2);
grid on;
xlabel('Steering Angle (Degrees)');
ylabel('Beamforming Gain (dB)');
title('Beamforming Gain vs Steering Angle');
axis([-90 90 -40 5]);
```

Output



Result:

The simulation shows that the beamforming gain reaches its maximum in the main beam direction and decreases as the steering angle moves away from the desired direction. Side lobes are observed at other angles with lower gain levels. This demonstrates that beamforming effectively focuses the transmitted or received signal toward a specific direction, thereby improving signal strength and reducing interference.

Question 5

Simulation of 5G Throughput Performance under QPSK, 16-QAM, 64-QAM, 256-QAM, and Varying SNR

Aim

To simulate the throughput performance of a 5G communication system using different modulation schemes (QPSK, 16-QAM, 64-QAM, and 256-QAM) under varying Signal-to-Noise Ratio (SNR) conditions.

MATLAB Code

```
clc;
clear;
close all;
% Parameters
SNR = 0:2:30;    % SNR in dB
BW = 100;        % 5G Bandwidth (MHz)
Efficiency = 0.9; % Coding Efficiency
% Spectral Efficiency (bits/s/Hz)
SE_QPSK = 2;
SE_16QAM = 4;
SE_64QAM = 6;
SE_256QAM = 8;
% Throughput Calculation
TP_QPSK = BW * Efficiency * log2(1 + 10.^(SNR/10)) * (SE_QPSK/2);
TP_16QAM = BW * Efficiency * log2(1 + 10.^(SNR/10)) * (SE_16QAM/2);
TP_64QAM = BW * Efficiency * log2(1 + 10.^(SNR/10)) * (SE_64QAM/2);
TP_256QAM = BW * Efficiency * log2(1 + 10.^(SNR/10)) * (SE_256QAM/2);
% Plot
figure;
```

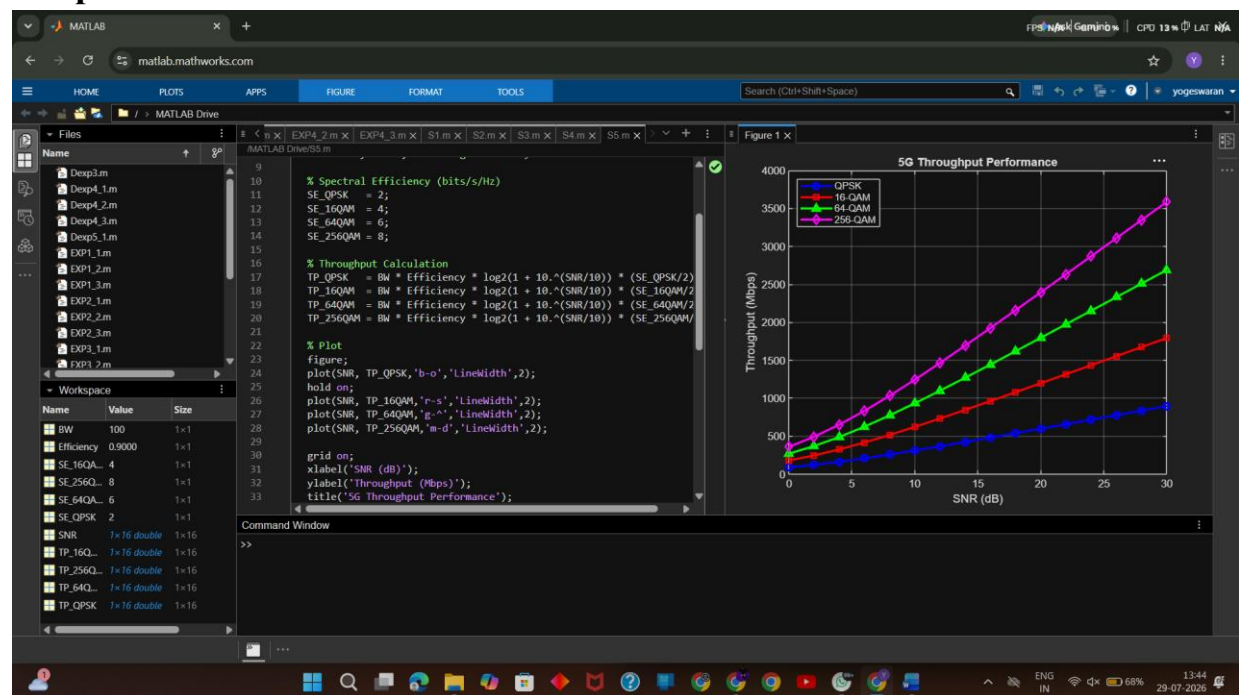
```

plot(SNR, TP_QPSK,'b-o','LineWidth',2);
hold on;
plot(SNR, TP_16QAM,'r-s','LineWidth',2);
plot(SNR, TP_64QAM,'g-^','LineWidth',2);
plot(SNR, TP_256QAM,'m-d','LineWidth',2);

grid on;
xlabel('SNR (dB)');
ylabel('Throughput (Mbps)');
title('5G Throughput Performance');
legend('QPSK','16-QAM','64-QAM','256-QAM','Location','northwest');

```

Output



Result:

The simulation shows that the throughput of the 5G communication system increases as the SNR increases for all modulation schemes. Among the modulation techniques, 256-QAM provides the highest throughput, followed by 64-QAM, 16-QAM, and QPSK. Higher-order modulation schemes improve spectral efficiency and data rate but require better channel conditions (higher SNR) for reliable communication.

